

Characterizing Methane Fluxes in Urban Heat Islands

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Abstract

This paper reports a systematic evaluation of basin-scale aerosol budgets. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The reduced model retains most of the explanatory power at a fraction of the computational cost. Sensitivity to the smoothing window proved modest across the range we examined.

1. Introduction

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Calibration drift over the trial period remained below the threshold set out in the protocol. The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting.

2. Related Work

Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters.

Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol.

Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

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3. Materials and Methods

Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Sensitivity to the smoothing window proved modest across the range we examined. Calibration drift over the trial period remained below the threshold set out in the protocol. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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4. Experimental Setup

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5. Results

We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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6. Discussion

Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Neither of the alternative formulations improved on the simple additive model in our setting. Taken together, these findings argue for treating the two regimes separately in future analyses. The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate.

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7. Limitations

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8. Conclusion

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